

# Sample-aware TCR join rebuild

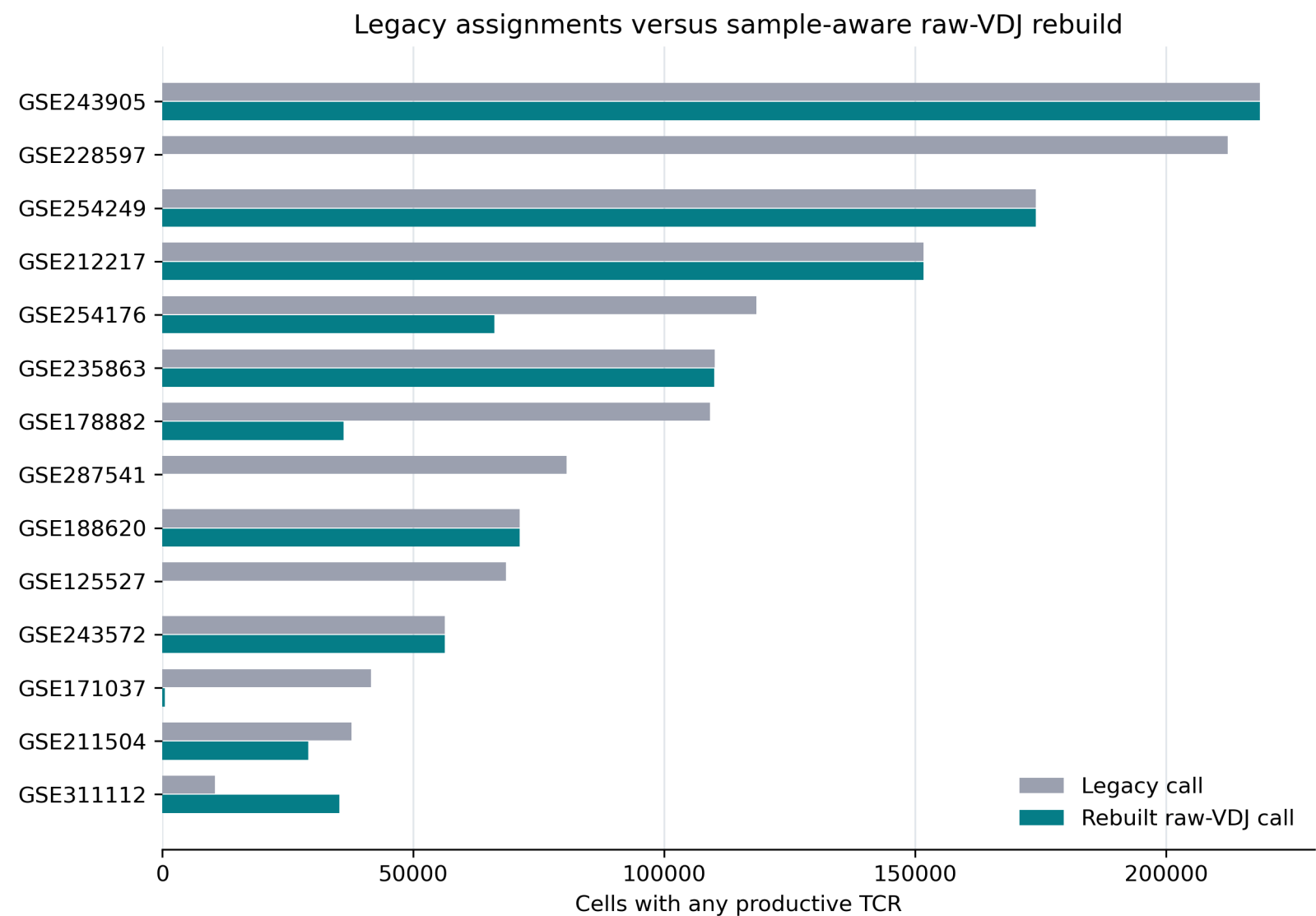
Productive raw VDJ contigs were rebuilt against RNA metadata using only `sample_id + barcode_core`. Per-chain UMI and read support are preserved for TRA, TRB, TRG, and TRD; missing support remains null.

|                                                                  |                                                                      |                                                                   |                                                                          |
|------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------|
| <div>11/14</div> <div>sources pass full or partial rebuild</div> | <div>2,479,137</div> <div>rows staged for metadata replacement</div> | <div>948,991</div> <div>cells with validated productive TCR</div> | <div>1,553,032</div> <div>productive chain calls with observed UMI</div> |
|------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------|

**QC gate:** the rebuild produced a checksum-bound sidecar and all 14 source H5AD size/mtime pairs remained unchanged.

**Three sources remain quarantined.** GSE125527 lacks recoverable RNA library identity; GSE287541 has no raw VDJ file; GSE228597 maps only 1 eligible raw cell key to RNA (0.0005%), below the frozen 50% source floor.

# Legacy assignments versus raw-VDJ rebuild



Red bars are diagnostic counts from quarantined sources and are not eligible for replacement or model truth.

# Rebuild method

## 1. Parse raw VDJ

Require cell-associated, high-confidence, productive contigs with nonempty CDR3.

## 2. Select within chain

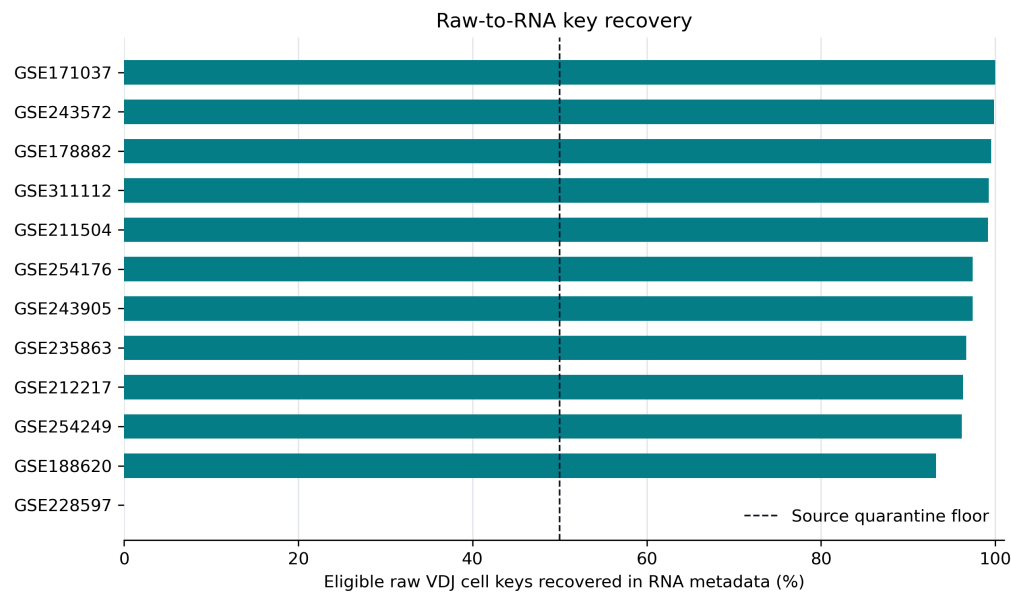
Highest UMI, then reads, full-length status, then stable contig id.

## 3. Join safely

Canonical sample id plus nucleotide barcode core. Barcode-only fallback is forbidden.

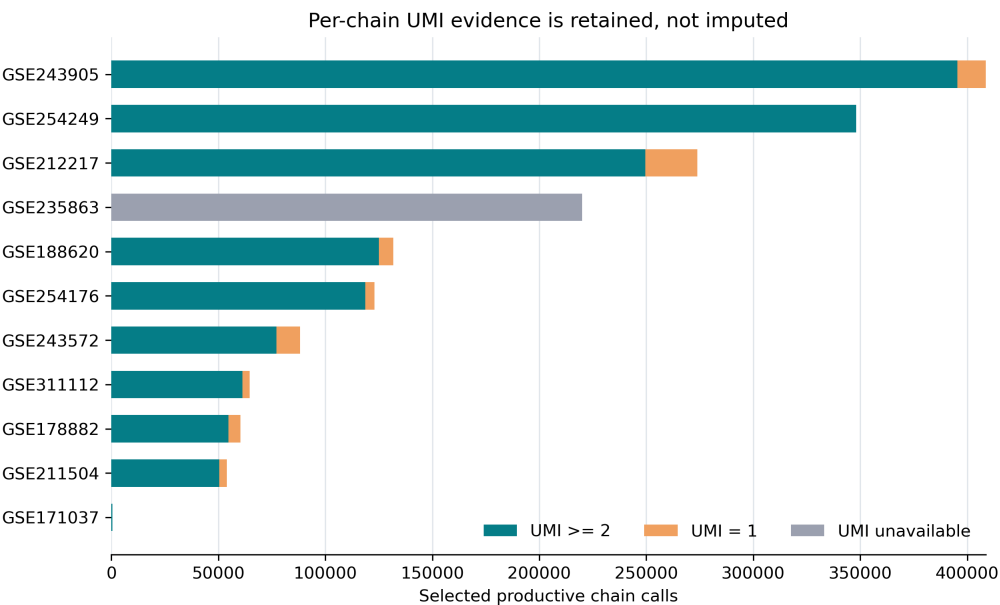
## 4. Fail closed

Duplicate RNA keys, collapsed raw libraries, or poor source recovery cannot provide TCR truth.



The 50% source floor only detects structural mapping failure; passing sources recover 93.2%-100.0% of eligible raw keys.

The four-chain sidecar contains CDR3 amino acid/nucleotide sequence, V/D/J calls, clone id, UMI, reads, support-availability flags, selected contig id, productive-contig multiplicity, and raw source-file provenance.



GSE235863 is an author receptor table without UMI/read columns. These are null, not zero. Across UMI-auditable calls, 72,049 selected chains have UMI=1.

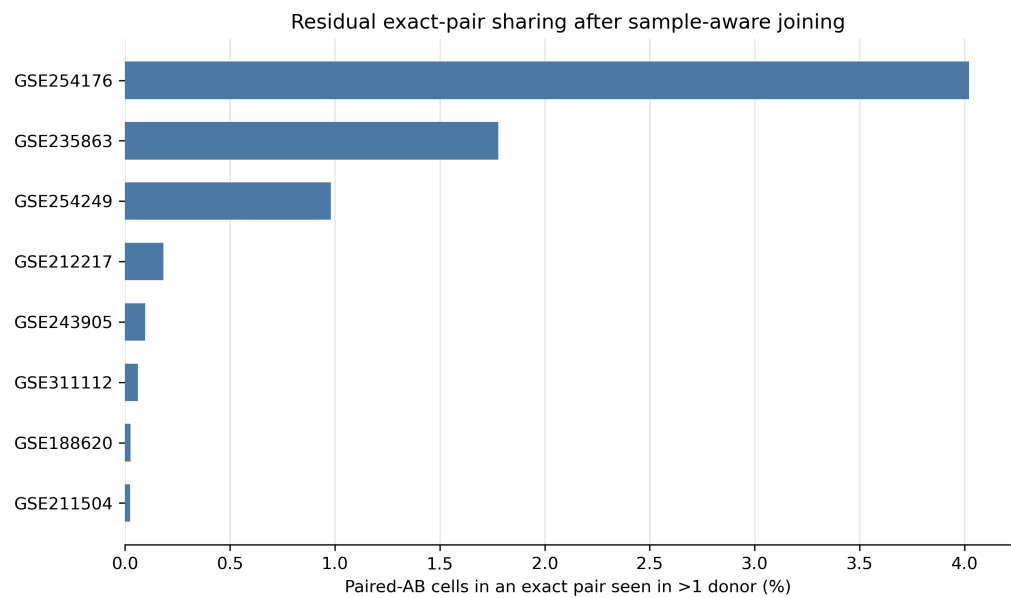
# Source-level outcome

| Source    | Outcome                    | Legacy any TCR | Rebuilt any TCR | Rebuilt paired AB | Raw-key recovery | Chain calls with UMI |
|-----------|----------------------------|----------------|-----------------|-------------------|------------------|----------------------|
| GSE125527 | Quarantined                | 68,439         | 0               | 0                 | NA               | 0                    |
| GSE171037 | Pass                       | 41,536         | 473             | 0                 | 100.0%           | 473                  |
| GSE178882 | Pass                       | 109,136        | 36,102          | 24,191            | 99.5%            | 60,293               |
| GSE188620 | Pass                       | 71,233         | 71,233          | 60,624            | 93.2%            | 131,857              |
| GSE211504 | Pass                       | 37,635         | 29,121          | 24,833            | 99.2%            | 53,954               |
| GSE212217 | Pass                       | 151,676        | 151,676         | 122,135           | 96.3%            | 273,811              |
| GSE228597 | Quarantined                | 212,272        | 0               | 0                 | 0.0%             | 0                    |
| GSE235863 | Pass, 110 rows quarantined | 110,081        | 109,985         | 109,985           | 96.7%            | 0                    |
| GSE243572 | Pass                       | 56,322         | 56,322          | 31,930            | 99.9%            | 88,252               |
| GSE243905 | Pass                       | 218,658        | 218,658         | 189,985           | 97.4%            | 408,643              |
| GSE254176 | Pass                       | 118,328        | 66,110          | 56,880            | 97.4%            | 122,990              |
| GSE254249 | Pass                       | 174,045        | 174,045         | 174,045           | 96.1%            | 348,090              |
| GSE287541 | Quarantined                | 80,545         | 0               | 0                 | NA               | 0                    |
| GSE311112 | Pass                       | 10,435         | 35,266          | 29,403            | 99.2%            | 64,669               |

**GSE311112 correction:** explicit support for relapse and BTK-clone sample names recovers 35,266 cells, compared with 9,663 under the incomplete historical parser.

GSE235863 has 110 duplicated RNA sample\_id + barcode\_core rows. Their rebuilt receptor fields are blank and they are excluded from receptor truth, but retained in the replacement table so stale assignments can be cleared.

# Residual biological and annotation checks



This sharing is already present in raw sample-resolved VDJ. It may include public or semi-invariant receptors; it is reported rather than used as automatic evidence of a join failure. Donor-level values are not reported for GSE178882 paired libraries or GSE243572 pooled libraries because donor identity is unresolved.

## Exact pair and author-label audit

| Source    | Paired AB | Cross-donor cells | Cross-donor % | Author NK | NK + rebuilt AB | NK + AB % |
|-----------|-----------|-------------------|---------------|-----------|-----------------|-----------|
| GSE125527 | 0         | 0                 | NA            | 5,471     | 0               | 0.00%     |
| GSE178882 | 24,191    | NA                | NA            | 0         | 0               | NA        |
| GSE188620 | 60,624    | 16                | 0.03%         | 0         | 0               | NA        |
| GSE211504 | 24,833    | 6                 | 0.02%         | 0         | 0               | NA        |
| GSE212217 | 122,135   | 223               | 0.18%         | 0         | 0               | NA        |
| GSE228597 | 0         | 0                 | NA            | 116,837   | 0               | 0.00%     |
| GSE235863 | 109,985   | 1,957             | 1.78%         | 24,676    | 0               | 0.00%     |
| GSE243572 | 31,930    | NA                | NA            | 0         | 0               | NA        |
| GSE243905 | 189,985   | 182               | 0.10%         | 501       | 245             | 48.90%    |
| GSE254176 | 56,880    | 2,287             | 4.02%         | 0         | 0               | NA        |
| GSE254249 | 174,045   | 1,707             | 0.98%         | 0         | 0               | NA        |
| GSE287541 | 0         | 0                 | NA            | 24,472    | 0               | 0.00%     |
| GSE311112 | 29,403    | 18                | 0.06%         | 0         | 0               | NA        |

**Interpretation:** raw productive TRA/TRB in an author-NK label is a conflict to review for NKT-like cells, doublets, or annotation error. It is not erased by transcriptomic NK labeling and is not automatically admitted as model truth.

# Staged metadata replacement

The staged sidecar contains all rows from the 11 passing sources. Receptor-truth eligibility is separate from metadata-replacement eligibility, allowing unsafe legacy calls to be cleared without converting ambiguous rows into negative training truth.

| Source    | Replacement rows | TCR-truth eligible | Validated productive TCR |
|-----------|------------------|--------------------|--------------------------|
| GSE171037 | 60,981           | 60,981             | 473                      |
| GSE178882 | 109,138          | 109,138            | 36,102                   |
| GSE188620 | 151,027          | 151,027            | 71,233                   |
| GSE211504 | 37,637           | 37,637             | 29,121                   |
| GSE212217 | 292,271          | 292,271            | 151,676                  |
| GSE235863 | 286,843          | 286,733            | 109,985                  |
| GSE243572 | 268,664          | 268,664            | 56,322                   |
| GSE243905 | 413,696          | 413,696            | 218,658                  |
| GSE254176 | 118,332          | 118,332            | 66,110                   |
| GSE254249 | 601,436          | 601,436            | 174,045                  |
| GSE311112 | 139,112          | 139,112            | 35,266                   |

**Artifact:** Integrated\_dataset/tables/tcr\_join\_rebuild/validated\_tcr\_replacement\_sidecar.parquet

**SHA-256:** 6a7450b4edb0ad2e4796a98e3bb9f9a6fffc059f723c34bcd2f260e65ddf253

**No propagation was performed.** Writing these fields into source or milestone H5AD metadata remains a high-risk mutation gate. The replacement should be applied atomically with backups only after explicit approval, followed by harmonized-metadata and milestone overlap checks.

This report evaluates data integrity of TCR assignment. It does not evaluate or promote a gdTAI classifier, threshold, or atlas inference.